R22

Code No.: 22CSC35

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. (Chem) III Sem (Main/Backlog) Examination December 2024

Data Structures using Python

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A

(5Q X 2M = 10 Marks)

		M	CO	BT
1	What is a class in Python?	(2)	1	2
2	Give the time complexity of merge sort.	(2)	2	2
3	Mention the advantages of a linked lists over an array.	(2)	3	4
4	Define a graph data structure in python?	(2)	4	1
5	Define a hash table and explain its basic operation.	(2)	5	4

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	Write a python program to create simple class and use it to instantiate objects in Python	(5)	1	3
	(b)	Define scatter plot and discuss its usefulness for data visualization with an example.	(5)	1	2
		(OR)			
7	(a)	Define a Pandas Series. Write a python program to create a Pandas Series. Also explain five member functions of Pandas Series.	(5)	1	2
	(b)	Write a Python program to group numeric data into bins using specific intervals?	(5)	1	3
8	(a)	Discuss in detail about asymptotic notation used for best case analysis of algorithms	(5)	2	3
	(b)	Perform selection Sort for the following list: 12, 3, 8, 23, 1, 8, 9, 34.	(5)	2	2
		(OR)			
9	(a)	What is data structure? What are the differences between linear and nonlinear data structures?	(5)	2	2
	(b)	Write a Python program to implement quicksort.	(5)	2	4
10	(a)	Write an algorithm to insert an element at the beginning of singly linked list.	(5)	3	1
	(b)	Write a Python program to implement a queue.	(5)	3	4
		(OR)			
11	(a)	Write the procedure to delete an element from the end of singly linked list.	(5)	3	1
	(b)	Write a Python program code snippet to display all the nodes of a doubly linked list in both forward and reverse order.	(5)	3	3

12	(a)	Define Binary Search tree? Draw a BST for the following List: 10,12,43,7,56,34,67,87,98 and delete 10 from the tree.	(5)	4	2
	(b)	List any five differences between breadth first search (BFS) and depth first search (DFS).	(5)	4	3
		(OR)			
13	(a)	Discuss briefly about any 3 types of binary trees.	(5)	4	2
	(b)	Explain graph representations in detail with examples	(5)	4	2
14		Discuss the Rabin- Karp string matching algorithm for given text and string, and analyse it's time complexity. Text: caccaadbafcd String: aadbaf	(10)	5	2
		(OR)			
15		Explain any three hash functions with examples and implement them using python code.	(10)	5	4

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